Notes: Bonfim, Ferreira, Queiró, Zhao (AER, 2025)

Fiscal Policy and Credit Supply in a Crisis

Contents

1 Big picture

- **Question:** Can *fiscal austerity* (cuts to public procurement) propagate through the banking system and depress *private* credit supply during a crisis?
- **Setting:** Portugal during the European sovereign debt crisis. The government cut public procurement by about **4.3% of GDP** (from peak to trough).
- **Core mechanism:** procurement cuts → government contractors lose revenues → contractors default more → banks get hit with contractor NPLs → bank balance sheets weaken → banks cut lending to *other (non-contractor) firms*.
- **Main empirical result:** A bank that is more exposed to procurement cuts reduces credit supply more. Estimated elasticity of *credit supply* w.r.t. procurement demand is about **2.5**.
- **Why it matters:** In a crisis, fiscal contractions can have *large, persistent* effects because they turn into a financial-sector shock (NPLs) and trigger a credit crunch.

2 Data and key objects

- **Public procurement contracts:** identify government contractors and their contract revenues by year.
- **Credit register (bank–firm loans):** universe of bank–firm credit exposures (Portugal).
- **Firm outcomes:** value added, sales, assets, employment, survival, etc.
- **Bank balance sheets:** equity, NPLs, sovereign debt holdings, recapitalizations.

Key variables.

- **Contractor-level procurement cut** (intensive margin) is defined as the drop in a contractor's procurement revenue from pre- to post-period:

$$\text{ProcurementCut}_i = \frac{1}{2}\sum_{t=2009}^{2010} \text{Procurement}_{it} - \frac{1}{5}\sum_{t=2011}^{2015} \text{Procurement}_{it}. \tag{2}$$

- **Bank-level procurement exposure** is a credit-share-weighted average of contractor cuts (scaled by contractor sales):

$$\text{ProcurementExposure}_b = \kappa \sum_{i=1}^{n} \left(\frac{\text{Credit}_{ib}}{\text{Credit}_b}\right)\left(\frac{\text{ProcurementCut}_i}{\text{Sales}_i}\right). \tag{1}$$

  Intuition: bank b is "more exposed" if it lent more (pre-crisis) to firms whose government business shrank more (as a share of their sales). The scalar κ is chosen so the *aggregate* procurement cut implied by the micro data matches the aggregate procurement cut in macro data.

- **Bank–firm credit growth outcome:** log cumulative credit growth in a relationship between 2010:IV and 2015:IV:

$$g_{ib} = \log\left(\frac{\sum_{t=2011:I}^{2015:IV} \text{Credit}_{ibt}}{\text{Credit}_{ib,2010:IV}}\right). \tag{3}$$

3 Empirical model(s)

3.1 Baseline credit-supply regression

They estimate:

$$g_{ib} = \beta \cdot \text{ProcurementExposure}_b + \gamma_{j(i)m(i)} + \lambda_1' X_b + \lambda_2' Z_i + \varepsilon_{ib}. \tag{4}$$

- $\gamma_{j(i)m(i)}$: firm **sector** $\times$ **municipality** fixed effects $\Rightarrow$ compare firms in the same sector and location (proxying for local demand/industry shocks).
- X_b: bank controls (size, equity, liquidity, etc.; plus robustness controls like sovereign exposure, construction exposure, etc.).
- Z_i: firm controls (size, age, baseline credit, etc.).
- **Identification logic:** conditional on sector×municipality FE and controls, variation in $\text{ProcurementExposure}_b$ is a bank-balance-sheet shock (coming from contractor distress), not loan demand by firm i.

Within-firm (multiple-bank) design. For firms borrowing from multiple banks, they also use firm fixed effects (Khwaja–Mian style): compare how *the same firm* is treated by banks with different procurement exposure.

Dynamics. For Figure 5 they run an event-study style version where the outcome is credit growth up to quarter t:

$$g_{ib,t} = \log\left(\frac{\text{Credit}_{ib,t}}{\text{Credit}_{ib,2010:IV}}\right), \qquad g_{ib,t} = \beta_t \text{ProcurementExposure}_b + \gamma_{j(i)m(i),t} + \dots$$

and plot $\{\beta_t\}$ over time (pre-trends should be near zero).

Firm-level outcomes. Define firm exposure as a credit-share-weighted average of its lenders' exposures (using pre-crisis loan shares):

$$\text{Exposure}_i \equiv \sum_b \omega_{ib} \, \text{ProcurementExposure}_b, \qquad \omega_{ib} = \frac{\text{Credit}_{ib,2010:IV}}{\sum_{b'} \text{Credit}_{ib',2010:IV}}.$$

Then regress firm-level outcomes (credit, survival, value added, etc.) on Exposure_i with sector×municipality FE and firm controls (Table 8; Figure 6).

4 Figures

4.1 Figure 1: the European sovereign debt crisis (macro motivation)

Interpretation / intuition. Countries with rising sovereign risk also experience persistent private credit contractions and weaker incomes. This motivates thinking about *feedback* between fiscal stress and bank credit supply.

4.2 Figure 2: procurement cuts are large in crisis countries

Interpretation. Procurement is quantitatively important in austerity packages. If procurement cuts transmit to banks via contractor defaults, they can be a major driver of credit contractions.

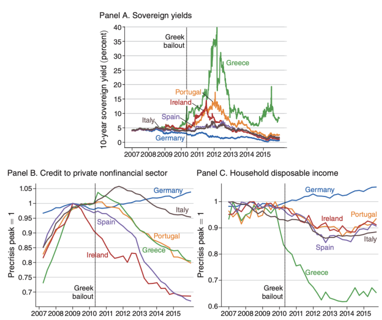

Figure 1. The European Sovereign Debt Crisis

Notes: Panel A plots ten-year sovereign yield data from LSEG (2022). Panel B plots credit data from ECB (2022), with the exception of credit from monetary and financial institutions in Greece, for which we use data from Bank of Greece (2022) that correct for a series break in June 2010. Panel C plots household income data from OECD (2022c). We present household income rather than GDP to exclude the effect of multinational corporations domiciled in Ireland for tax reasons (see OECD 2016 for a discussion of this issue).

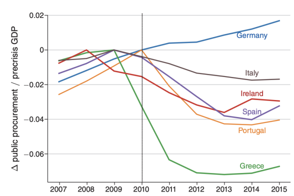

Figure 2. Public Procurement in the Crisis

Note: This figure plots the change in real public procurement spending relative to its precrisis peak, as a fraction of precrisis GDP. Data are from OECD (2022a, b).

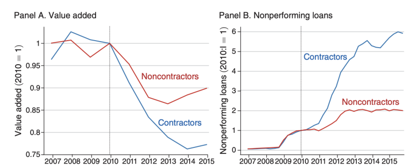

Figure 3. Impact of Procurement Cuts on Firms

Note: Panels A and B plot the evolution of value added and NPLs, respectively, for firms with public procurement contracts in 2009–2010 (contractors) versus firms without such contracts in 2009–2010 (noncontractors).

4.3 Figure 3: contractors are hit harder, and default more

Interpretation. Procurement cuts show up as a large negative shock to contractor cashflows (value added) and a strong increase in loan distress. This is the "upstream" shock feeding into banks.

4.4 Figure 4: procurement-driven NPL losses are persistent vs. sovereign valuation losses

Interpretation. The procurement shock creates *credit losses* that persist on bank balance sheets. This makes it plausible that procurement cuts can generate a long-lived credit supply contraction, even after sovereign spreads normalize.

4.5 Figure 5: core credit-supply result and dynamics

Interpretation. No evidence of pre-trends (supports identification). The negative effect grows over time, consistent with *slow-moving* balance-sheet repair: contractor NPLs accumulate and tie up bank capital for years.

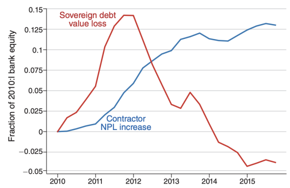

Figure 4. Impact of the Procurement and Sovereign Debt Shocks

Notes: This figure plots the increase in NPLs from firms with public procurement contracts in 2009–2010 (contractors) versus the loss in the market value of bank domestic sovereign debt holdings. Both series are plotted as a fraction of total bank equity in 2010:I. Our estimate for the change in the aggregate market value of domestic sovereign debt is based on data on debt holdings from BPLIM (2021), the average residual maturity from the EBA's 2011 stress test data (EBA 2011b), sovereign yield data from LSEG (2022), and the average interest rate on outstanding debt in 2010 reported by IGCP (2018).

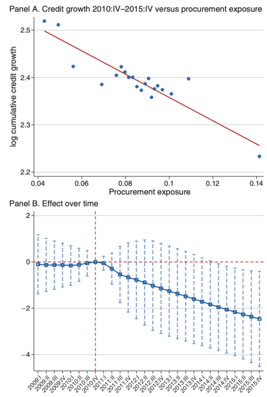

Figure 5. Effect of Procurement Exposure on Credit at the Bank-Firm Level

Notes: Panel A presents a binned scatterplot of the log of cumulative credit growth between 2010:IV and 2015:IV versus procurement exposure at the bank-firm level. Panel B shows point estimates and 95 percent confidence intervals for the bank-firm-level effect of procurement exposure on log cumulative credit growth between 2010:IV and each quarter between 2009:I and 2015:IV.

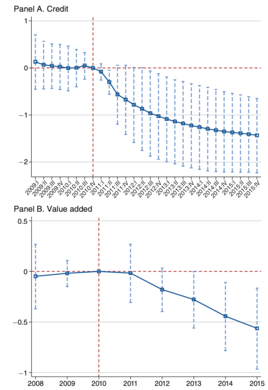

Figure 6. Effect of Procurement Exposure at the Firm Level

Notes: Panel A shows point estimates and 95 percent confidence intervals for the firm-level effect of procurement exposure on log cumulative credit growth between 2010:IV and each quarter between 2009:I and 2015:IV. Panel B presents estimates for log cumulative value added growth between 2010 and each year between 2008 and 2015.

4.6 Figure 6: firm-level credit and real effects

Interpretation. The bank-level credit supply shock passes through to firms' total borrowing, and then to real activity (value added). The lag in real outcomes is consistent with financial constraints biting gradually (rollovers, investment, working capital).

5 Tables (what each one is showing)

5.1 Tables 1–2: summary statistics and balancing

Read.

- Procurement exposure has mean about 0.085 (bank–firm sample), i.e. economically meaningful heterogeneity across banks.
- Balance: high-exposure banks are not obviously "worse" on most observables, but they do have (i) higher pre-crisis NPL ratios and (ii) more construction exposure. Importantly, controlling for these does not kill the main effect (Table 3).
- This table is basically: "procurement exposure is not just a proxy for sovereign exposure or bank size."

5.2 Table 3: baseline evidence that procurement exposure contracts credit supply

Key coefficient (col. 1): $\hat{\beta} \approx -2.460$ (s.e. 0.682).

- Because the outcome is in logs, **a 1 percentage point increase** in procurement exposure (+0.01) implies about 2.46% lower cumulative credit in that bank–firm relationship.
- At the **mean exposure** (0.085), the implied log change is $-2.46 \times 0.085 \approx -0.209$, i.e. roughly $\exp(-0.209) - 1 \approx -19\%$ lower cumulative credit supplied.

Table 1—Sample Summary Statistics

	Mean	SD	P10	Median	P90
Panel A. Bank-firm matched sample					
Bank-firm variables					
Total credit (€ thousand)	262.87	446.14	31.82	94.94	651.44
Bank variables					
Procurement exposure	0.08	0.02	0.05	0.09	0.11
Sovereign debt exposure	0.64	0.59	0.11	0.40	1.72
Total assets (€ billion)	64.23	38.68	17.57	60.43	112.46
Equity-to-assets	0.08	0.03	0.05	0.08	0.11
Liquidity	0.04	0.01	0.02	0.04	0.06
Foreign bank	0.19	0.39	0.00	0.00	1.00
Credit/assets	0.69	0.06	0.65	0.66	0.76
NPL/total credit	0.10	0.06	0.03	0.09	0.16
Observations	76,289				
Panel B. Firm-level sample					
Firm variables					
Value added (€ thousand)	312.88	479.61	32.92	137.23	758.30
Sales (€ thousand)	1,272.50	2,185.07	102.25	451.96	3,179.89
Total assets (€ thousand)	1,392.01	2,456.17	126.52	499.00	3,406.39
Cash (€ thousand)	81.19	146.12	1.64	25.61	213.56
Employment	12.60	16.65	2.00	7.00	30.00
Return on assets	−0.02	5.03	−0.08	0.05	0.16
Leverage	0.40	0.41	0.11	0.35	0.71
Current ratio	0.64	0.37	0.21	0.70	0.97
Bank variables					
Procurement exposure	0.08	0.02	0.05	0.09	0.11
Sovereign debt exposure	0.62	0.52	0.11	0.42	1.62
Total assets (€ billion)	62.65	34.16	17.57	60.43	111.25
Equity-to-assets	0.08	0.02	0.05	0.08	0.10
Liquidity	0.04	0.01	0.03	0.04	0.06
Foreign bank	0.17	0.31	0.00	0.00	0.79
Credit/assets	0.69	0.05	0.64	0.67	0.76
NPL/total credit	0.10	0.05	0.04	0.09	0.16
Observations	50,346				

Notes: This table reports summary statistics for the bank-firm matched sample (panel A) and firm-level sample (panel B). Bank-level variables are measured in 2010:I and firm-level variables in 2010. In panel B, bank variables are aggregated by firm using the credit shares of each bank as weights.

Table 2—Balancing of Bank Covariates

	Procurement exposure			
	Below median	Above median	Difference	SE
Sovereign debt exposure	0.706	0.635	0.071	(0.351)
Total assets (log)	9.944	10.229	−0.284	(0.671)
Equity-to-assets	0.067	0.067	−0.001	(0.019)
Liquidity	0.041	0.034	0.007	(0.009)
Foreign bank	0.500	0.286	0.214	(0.290)
Credit/assets	0.738	0.745	−0.007	(0.061)
NPL/total credit	0.051	0.139	−0.089	(0.036)
Construction exposure	0.156	0.227	−0.071	(0.033)
Predicted growth in other NPLs	0.084	0.079	0.005	(0.008)
Recapitalized	0.333	0.571	−0.238	(0.292)
Predicted growth in other credit				
By financing type	−0.120	−0.112	−0.008	(0.033)
By collateral type	0.000	0.006	−0.005	(0.038)
By sector	−0.114	−0.124	0.011	(0.025)
By location	−0.102	−0.097	−0.005	(0.011)

Notes: This table compares precrisis (2010:I) covariates of banks with below- and above-median procurement exposure. The table reports means, the difference in means, and the standard error of the difference in means for each group of banks. Variable definitions are provided in Table A1 in the Appendix.

Table 3—Effect of Procurement Exposure on Bank-Firm-Level Credit

			Controls for other credit supply shocks		
	Baseline	Survival	Construction exposure	Predicted growth in other NPLs	Recapitalizations
	(1)	(2)	(3)	(4)	(5)
Procurement exposure	−2.460	−1.522	−2.578	−2.597	−2.557
	(0.682)	(0.766)	(0.672)	(0.821)	(0.759)
BM degrees of freedom	3.3	3.3	3.4	3.3	3.2
Observations	76,289	76,289	76,289	76,289	76,289
Adjusted R^2	0.067	0.102	0.068	0.069	0.067

Notes: This table presents estimates from credit regressions using bank-firm matched data. The dependent variable is the log cumulative growth in credit between 2010:IV and 2015:IV, except in column 2, where it is an indicator for whether a relationship survived until 2015:IV. All regressions control for precrisis sovereign debt exposure, total assets, and the equity-to-assets ratio at the bank level, as well as for precrisis log total assets, return on assets, leverage, and the current ratio at the firm level. Column 3 adds the share of credit to the construction sector in 2010:I to the set of bank controls. Column 4 adds a shift-share predictor of NPL growth for noncontractors during the crisis, in which the shares are bank exposures by sector in 2010:I and the shifters are the leave-one-out national changes in NPLs as a share of precrisis credit in each sector between 2010:I and 2015:IV. Column 5 adds an indicator for whether a bank was recapitalized. Standard errors in parentheses are clustered at the bank level using the "LZ2" bias-reduction modification of Imbens and Kolesár (2016). The "BM degrees of freedom" row reports the degrees of freedom suggested by Bell and McCaffrey (2002) to compute t-distribution confidence intervals for the coefficient on procurement exposure.

- Cols. (3)–(5): adding construction exposure, predicted other-NPL growth, and recapitalization controls leaves $\hat{\beta}$ essentially unchanged $\Rightarrow$ not a generic "bad bank" story.

5.3 Table 4: where does identification come from? (Rotemberg weights)

Interpretation.

- The exposure measure is a shift-share object. This table checks which "shares×shifters" drive identification.
- **Construction dominates:** about 84.5% of the weight comes from construction contractors. Moreover, within construction, a tiny set of large contractors (top 5%) carry most of the weight.
- Takeaway: the core quasi-experiment is largely about procurement cuts in construction and banks that pre-lent to those big contractors. This is both a strength (clear shock) and a caution (external validity depends on how representative construction is).

5.4 Table 5: alternative ways to address credit demand

Interpretation.

- Multi-bank firms: even with **firm fixed effects**, the exposure coefficient remains large (≈ -2.59).
- Predicted-demand controls: whether I predict demand via financing type, collateral, sector, or location, $\hat{\beta}$ stays around -2.2 to -2.4.
- Overall: the evidence strongly supports a **credit supply** interpretation.

5.5 Table 6: amplification via bank leverage and mitigation via recapitalization

Interpretation.

- The baseline effect becomes more negative (around -4.3 to -4.7) when interacting with capitalization, consistent with a **balance-sheet constraint**.

- Interaction with equity-to-assets is positive: better-capitalized banks are less sensitive to the procurement-induced NPL shock.
- Recapitalization matters: procurement exposure × recapitalized is strongly positive ($\approx +3.12$), meaning recapitalized banks attenuate the credit contraction driven by procurement exposure.

TABLE 4—DECOMPOSING THE EFFECT OF EXPOSURE

	$\hat{\alpha}_j$ (1)	Top 5% share (2)	$\hat{\beta}_j$ (3)
Construction	0.845	0.811	−2.04
Administrative services	0.039	0.738	−2.82
Water and waste management	0.032	0.957	−2.50
Consulting	0.029	0.546	−6.86
Wholesale and retail trade	0.018	0.466	−3.61

Notes: This table lists the top five sectors by the sum of Rotemberg weights ($\hat{\alpha}_j$), calculated following the decomposition proposed by Goldsmith-Pinkham, Sorkin, and Swift (2020). Column 2 displays the fraction of weights within each sector accounted for by the top 5 percent of contractors by weight. Column 3 reports the weighted average coefficient on exposure obtained from the decomposition for each sector.

TABLE 5—ALTERNATIVE CONTROLS FOR CREDIT DEMAND

	Firms with multiple relationships		Controls for predicted growth in other credit			
	Baseline (1)	Within firm (2)	Financing type (3)	Collateral type (4)	Sector (5)	Location (6)
Procurement exposure	−2.306	−2.593	−2.444	−2.420	−2.196	−2.412
	(0.898)	(0.810)	(0.727)	(0.521)	(0.689)	(0.746)
BM degrees of freedom	3.3	5.1	3.2	3.3	3.3	3.7
Observations	41,138	41,138	76,289	76,289	76,289	76,289
Adjusted R^2	0.099	0.297	0.068	0.070	0.070	0.067

Notes: This table presents estimates from credit regressions using bank-firm matched data. The dependent variable is the log cumulative growth in credit between 2010:IV and 2015:IV. All regressions control for precrisis sovereign debt exposure, total assets, and the equity-to-assets ratio at the bank level, as well as for precrisis log total assets, return on assets, leverage, and the current ratio at the firm level. Columns 1 and 2 restrict the sample to firms with at least two lending relationships, and column 2 includes firm fixed effects. Column 3 adds a shift-share predictor of credit growth for noncontractors during the crisis, where the shares are bank exposures by financing type in 2010:I and the shifters are the leave-one-out national credit growth rates for each financing type between 2010:I and 2015:IV. Columns 4, 5, and 6 add analogous predictors of credit growth based on precrisis exposures to credit collateral types, sectors and municipalities, respectively. Standard errors in parentheses are clustered at the bank level using the "LZ2" bias-reduction modification of Imbens and Kolesár (2016). The "BM degrees of freedom" row reports the degrees of freedom suggested by Bell and McCaffrey (2002) to compute *t*-distribution confidence intervals for the coefficient on procurement exposure.

TABLE 6—INTERACTION WITH BANK LEVERAGE AND RECAPITALIZATIONS

	Equity-to-assets (1)	Recapitalizations (2)	Equity-to-assets with public injections (3)	Equity-to-assets with all injections (4)
Procurement exposure	−4.276	−4.515	−4.668	−4.716
	(0.990)	(0.852)	(0.819)	(0.660)
Equity-to-assets	−0.366		−0.403	−0.439
	(0.274)		(0.300)	(0.259)
Procurement exposure × equity-to-assets	28.180		33.843	34.640
	(18.643)		(18.537)	(15.157)
Recapitalized		−0.028		
		(0.006)		
Procurement exposure × recapitalized		3.121		
		(0.535)		
BM degrees of freedom for interaction	3.8	2.9	4.1	4.7
Observations	76,289	72,648	76,289	76,289
Adjusted R^2	0.068	0.069	0.068	0.069

Notes: This table presents estimates from credit regressions using bank-firm matched data. The dependent variable is the log cumulative growth in credit between 2010:IV and 2015:IV. All regressions control for precrisis sovereign debt exposure, total assets, and the equity-to-assets ratio at the bank level, as well as for precrisis log total assets, return on assets, leverage, and the current ratio at the firm level. In column 2, "Recapitalized" is an indicator for whether a bank was recapitalized in the 2010–2013 period. In column 3, the equity-to-assets ratio includes the capital injections banks received from the government in 2010–2013. In column 4, the equity-to-assets ratio additionally includes private capital injections in the same period. Standard errors in parentheses are clustered at the bank level using the "LZ2" bias-reduction modification of Imbens and Kolesár (2016). The "BM degrees of freedom" row reports the degrees of freedom suggested by Bell and McCaffrey (2002) to compute *t*-distribution confidence intervals for the coefficient on procurement exposure.

5.6 Table 7: effects across firm types

Interpretation. The effect is negative across all groups:

- by risk: $\hat{\beta} \approx -2.44$ (low risk) and -2.59 (high risk),
- by size: -2.28 (small) and -2.74 (large),
- by age: -2.50 (young) and -2.45 (old).

So the procurement-induced bank shock is **broad-based**, not only a "small risky firms" story.

5.7 Table 8: firm-level real effects (credit, survival, activity)

Interpretation. Higher exposure implies worse outcomes:

- total credit: $\hat{\beta} \approx -1.431$ (s.e. 0.293),
- survival probability: $\hat{\beta} \approx -0.400$ (0.094),
- value added: $\hat{\beta} \approx -0.563$ (0.148),
- sales: $\hat{\beta} \approx -0.623$ (0.218),
- assets: $\hat{\beta} \approx -0.317$ (0.075),
- employment: $\hat{\beta} \approx -0.410$ (0.134).

This is the "real economy" punchline: bank balance-sheet damage from fiscal austerity has material effects on firm activity.

TABLE 7—HETEROGENEOUS EFFECTS

	Credit risk		Firm size		Firm age	
	Low	High	Small	Large	Young	Old
	(1)	(2)	(3)	(4)	(5)	(6)
Procurement exposure	−2.435	−2.587	−2.277	−2.735	−2.500	−2.450
	(0.805)	(0.644)	(0.511)	(0.866)	(0.573)	(0.797)
BM degrees of freedom	3.4	3.7	3.6	3.4	3.5	3.5
Observations	40,913	33,400	41,050	33,413	42,596	31,864
Adjusted R^2	0.077	0.087	0.080	0.076	0.083	0.077

Notes: This table presents estimates from credit regressions using bank-firm matched data. The dependent variable is the log cumulative growth in credit between 2010:IV and 2015:IV. All regressions control for precrisis sovereign debt exposure, total assets, and the equity-to-assets ratio at the bank level, as well as for precrisis log total assets, return on assets, leverage, and the current ratio at the firm level. Split samples are defined using median values in 2010. Credit risk is the probability of default from SIAC, a credit assessment system developed by Banco de Portugal to provide individual credit risk ratings to enterprises. Firm size equals total assets. Firm age is calculated from the date of incorporation. Standard errors in parentheses are clustered at the bank level using the "LZ2" bias-reduction modification of Imbens and Kolesár (2016). The "BM degrees of freedom row" reports the degrees of freedom suggested by Bell and McCaffrey (2002) to compute t-distribution confidence intervals for the coefficient on procurement exposure.

TABLE 8—EFFECT OF PROCUREMENT EXPOSURE ON FIRM-LEVEL OUTCOMES

	Credit	Survival	Value added	Sales	Assets	Employment
	(1)	(2)	(3)	(4)	(5)	(6)
Procurement exposure	−1.431	−0.400	−0.563	−0.623	−0.317	−0.410
	(0.293)	(0.094)	(0.148)	(0.218)	(0.075)	(0.134)
BM degrees of freedom	4.3	4.3	4.3	4.3	4.3	4.3
Observations	50,346	50,346	50,346	50,346	50,346	50,346
Adjusted R^2	0.087	0.108	0.277	0.239	0.224	0.172

Notes: This table presents estimates from regressions for firm-level outcomes. The dependent variable is the log cumulative growth in each outcome between 2010 and 2015 (2010:IV and 2015:IV for credit), except in column 2, where it is an indicator for whether the firm survived until 2015. All regressions control for precrisis sovereign debt exposure, total assets, and the equity-to-assets ratio at the bank level, as well as for precrisis log total assets, return on assets, leverage, and the current ratio at the firm level. Procurement exposure and bank controls are aggregated to the firm level using the credit shares of each bank as weights. Standard errors in parentheses are clustered at the level of the main bank by loan size, using the "LZ2" bias-reduction modification of Imbens and Kolesár (2016). The "BM degrees of freedom" row reports the degrees of freedom suggested by Bell and McCaffrey (2002) to compute t-distribution confidence intervals for the coefficient on procurement exposure.

5.8 Table 9: calibration targets and parameters in the model

Interpretation.

- $\bar{v}$ (bank leverage) is pinned down by data (≈ 12.86).
- θ (elasticity of substitution across banks / loan-demand elasticity) is estimated from loan growth vs. lending rate changes (≈ 4.55).
- η (elasticity of loan supply to the interest rate, i.e. how easily banks can scale lending by raising funding) is calibrated (≈ 0.492) to match bank-level moment(s).
- Default probability ρ is chosen to match the empirical $\hat{\beta}$ through the model mapping $\beta = -\rho\bar{v}\frac{\theta}{\theta+\eta}$ (see below), giving $\rho \approx 0.212$.

5.9 Table 10: aggregate elasticities and counterfactuals

Interpretation.

- Baseline (crisis): aggregate credit elasticity $\varepsilon^Q \approx 2.13$ and output elasticity $\varepsilon^Y \approx 0.91$ — **large amplification**.
- Higher bank leverage amplifies strongly: $\bar{v} = 20 \Rightarrow \varepsilon^Q \approx 3.31$ and $\varepsilon^Y \approx 1.42$.
- "No crisis" (banks can elastically expand balance sheets, η much larger): elasticities drop sharply (e.g. $\varepsilon^Q \approx 0.66$, $\varepsilon^Y \approx 0.28$ at baseline leverage).
- Bottom line: the **same fiscal procurement cut** can have vastly different macro effects depending on bank leverage and the state of financial conditions.

TABLE 9—CALIBRATION

Parameter	Description	Value	Source/Target
$1/\phi$	Frisch elasticity	0.750	Herreño (2023), based on Chetty et al. (2011)
$\bar{v}$	Initial bank leverage	12.860	Aggregate bank leverage ratio in 2010:IV
θ	Elasticity of substitution across banks	4.550	Elasticity of credit demand to loan rates (column 4 of Table B.10 in Supplemental Appendix)
η	Elasticity of credit supply to loan rates	0.492	Elasticity of credit supply to bank net worth (column 2 of Table B.11 in Supplemental Appendix)
ρ	Probability of default	0.212	Elasticity of credit supply to procurement demand (column 1 of Table 3)

TABLE 10—AGGREGATE EFFECTS OF THE SHOCK IN THE MODEL

	Calibrations			Counterfactuals				
		Using β_{firm}				No crisis $\eta = 5.452$		
	Baseline	$\alpha = 1$	$\alpha = 1{,}000$	$\bar{v} = 8$	$\bar{v} = 20$	Baseline $\bar{v}$	$\bar{v} = 8$	$\bar{v} = 20$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ε^Q	2.128	2.330	1.915	1.324	3.309	0.662	0.412	1.030
ε^Y	0.912	0.999	0.821	0.567	1.418	0.284	0.177	0.442

Notes: Column 1 presents results from our baseline calibration. Columns 2 and 3 correspond to the alternative calibrations using the firm-level credit regression. The remaining columns report results for counterfactuals. In columns 4 and 5, we vary the level of initial bank leverage $\bar{v}$. In columns 6, 7, and 8, we use the elasticity of credit supply to lending rates η estimated in a noncrisis context, along with variation in $\bar{v}$.

6 Structural model (all the key equations)

The model rationalizes why a procurement shock becomes a bank credit supply shock, and how the effect is amplified in crises.

6.1 Preferences and aggregation

Representative household utility (quasi-linear):

$$U(C,L) = C - \frac{L^{1+\phi}}{1+\phi}. \tag{5}$$

CES consumption aggregator across differentiated goods:

$$C = \left(\int_0^1 c_i^{\frac{\sigma-1}{\sigma}}\, di\right)^{\frac{\sigma}{\sigma-1}}. \tag{6}$$

Labor aggregator (with elasticity parameter α) and implied wage indices:

$$L = \left(\int_0^1 l_i^{\frac{\alpha+1}{\alpha}}\, di\right)^{\frac{\alpha}{\alpha+1}}, \tag{7}$$

$$w = \left(\int_0^1 w_i^{\alpha+1}\, di\right)^{\frac{1}{\alpha+1}}. \tag{8}$$

Government procurement is the same CES bundle:

$$G = \left(\int_0^1 g_i^{\frac{\sigma-1}{\sigma}}\, di\right)^{\frac{\sigma}{\sigma-1}}. \tag{9}$$

6.2 Firms and borrowing

Firms produce with linear technology $y_i = z_i l_i$ and face CES demand $y_i = Y p_i^{-\sigma}$, where $Y = C + G$. They must **borrow** to finance wage bills from banks. Firm i's loan demand aggregates across banks:

$$q_i = \left(\int_0^1 \omega_{ib}^{1/\theta}\, q_{ib}^{\frac{\theta-1}{\theta}}\, db\right)^{\frac{\theta}{\theta-1}}. \tag{10}$$

Optimal allocation across banks (CES demand):

$$q_{ib} = q_i\, \omega_{ib} \left(\frac{R_b}{R_i}\right)^{-\theta}, \qquad R_i = \left(\int_0^1 \omega_{ib}\, R_b^{1-\theta}\, db\right)^{\frac{1}{1-\theta}}. \tag{11–12}$$

6.3 Banks

Banks have market power (markup) and face convex costs of leverage/capacity. Optimal lending rate:

$$R_b = \frac{\theta}{\theta-1}\, v_b^{1/\eta}. \tag{13}$$

Balance sheet (leverage constraint):

$$v_b\, e_b = q_b, \qquad q_b \equiv \int_0^1 q_{ib}\, di. \tag{14}$$

6.4 How procurement cuts hit bank equity

Let u_b be the model analog of a bank's exposure to the procurement shock (a loan-share-weighted average of the procurement contraction across its borrowers). If procurement cuts induce default with probability ρ, then:

$$\frac{de_b}{e_b} = -\rho\, v_b\, u_b. \tag{15}$$

Interpretation: higher leverage v_b amplifies losses from a given borrower-side shock u_b; this directly captures why crises (high leverage, thin capital) are dangerous.

6.5 Mapping to the empirical β

Proposition 1 (bank-level credit supply elasticity). The elasticity of bank lending to the procurement shock is:

$$\beta = -\rho\, \bar{v}\, \frac{\theta}{\theta + \eta}. \tag{16}$$

Reading: the empirical $\hat{\beta}$ pins down how default risk (ρ), bank leverage ($\bar{v}$), and the ability to adjust lending rates / balance sheets (θ, η) combine to produce a credit supply contraction.

6.6 Aggregate amplification

Proposition 2 (aggregate credit and output elasticities). Let $\psi \equiv 1/\phi + 1$ be the (absolute) elasticity of aggregate loan demand to interest rates. Then aggregate credit elasticity is:

$$\varepsilon_Q = -\frac{\psi}{\theta} \cdot \frac{\theta + \eta}{\psi + \eta} \cdot \beta, \tag{17}$$

and output elasticity is:

$$\varepsilon_Y = \frac{1}{1 + \phi} \varepsilon_Q. \tag{18}$$

So a micro credit-supply estimate (β) maps into macro effects through (i) demand elasticities and (ii) labor supply/Frisch elasticity.

6.7 Estimating θ (loan-demand elasticity across banks)

They estimate θ from:

$$\log\left(\frac{\text{Credit}_{ib,t}}{\text{Credit}_{ib,2010:IV}}\right) = \theta\, \log\left(\frac{R_{b,t}}{R_{b,2010:IV}}\right) + \gamma_{j(i)m(i),t} + \varepsilon_{ib,t}. \tag{19}$$

with sector×municipality×time fixed effects to absorb firm-side demand changes.

7 Conclusion / intuition

Fiscal austerity can be much more than a direct demand contraction: in Portugal's crisis, cutting procurement reduced contractors' revenues and sharply increased their loan defaults, which damaged banks' balance sheets and led exposed banks to cut lending to unrelated firms for years. The key intuition is that a procurement cut becomes an *endogenous bank-capital shock*: when banks are highly levered and cannot easily raise new funding (crisis conditions), even a sectoral fiscal cut can trigger broad credit supply reductions and sizable real effects. Recapitalizations dampen this amplification, and a calibrated general equilibrium model implies that the aggregate credit/output elasticities to procurement demand are large in crises but much smaller in normal times.

8 Conclusion

8.1 Summary

- **Research question.** Can fiscal austerity propagate through the banking system and depress *private* credit supply in a crisis, beyond standard aggregate-demand channels?
- **Setting and shock.** In Portugal during the sovereign-debt crisis, the government sharply cut public procurement (about **4.3% of GDP** from peak to trough), generating a large negative revenue shock for government contractors.
- **Key mechanism (the paper's causal chain).**

  Procurement cuts $\Rightarrow$ contractor revenue losses $\Rightarrow$ contractor loan defaults (NPLs) $\Rightarrow$ bank balance-sheet

  The distinctive feature is the *spillover*: a fiscal contraction aimed at the public sector becomes a broad private-credit shock via banks.
- **Empirical design.** The paper builds a bank-level exposure measure by combining: (i) contractor-specific procurement-revenue cuts and (ii) pre-crisis bank lending shares to those contractors. Credit supply is then estimated in bank–firm data with rich fixed effects (sector×municipality and time), and validated using within-firm (multi-bank) comparisons.
- **Main credit-supply result.** Banks more exposed to procurement-hit contractors cut lending significantly more to *non-contractors*. Quantitatively, the baseline estimate is about

  $$\hat{\beta} \approx -2.46,$$

  meaning a **1 percentage point** increase in exposure (+0.01) predicts roughly **2.46%** lower cumulative bank–firm credit over the post-crisis window; at the mean exposure, implied credit is about **19%** lower.
- **Balance-sheet amplification and policy mitigation.** The contraction is stronger for weaker (more constrained) banks and is *attenuated* by recapitalization, consistent with a bank-capital channel rather than a generic demand story.

- **Real effects.** Firms more exposed to procurement-hit banks experience materially worse outcomes: less total credit, lower survival probabilities, and declines in value added, sales, assets, and employment.

8.2 Economic intuition: the simple way to explain the result

- **A procurement cut is not "just" a demand shock in a crisis.** In normal times, a public-demand reduction would mainly work through standard expenditure multipliers. In a crisis, however, contractors' cash-flow losses become *credit losses for banks.*

- **Why the credit spillover is large in crises.** When banks are highly levered and equity/funding is hard to raise, a rise in NPLs forces balance-sheet repair. Repair is achieved largely by shrinking assets (credit), so lending falls even to firms unrelated to procurement.

- **Why recapitalization matters.** Injecting capital relaxes the binding balance-sheet constraint, allowing banks to absorb contractor defaults with less deleveraging, and therefore weakens the transmission from fiscal austerity to private credit supply.

8.3 Model-based conclusion (why the paper goes beyond reduced form)

- The paper embeds the empirics in a general-equilibrium banking framework that maps the reduced-form credit-supply estimate into structural parameters (loan-demand elasticity, bank leverage, and the elasticity of bank balance-sheet expansion).

- In the calibrated crisis environment, the model implies **large aggregate amplification**: procurement demand shocks generate sizable elasticities of aggregate *credit* and *output.*

- In counterfactual "no crisis" environments where banks can expand balance sheets more elastically, the same procurement cut produces **much smaller** aggregate effects.

- Bottom line: **state dependence** is central—the fiscal-to-credit multiplier is much larger when the banking system is constrained.

8.4 Takeaway

In Portugal's crisis, procurement austerity propagated through banks: cutting public contracts raised contractor defaults, weakened bank balance sheets, and induced exposed banks to cut lending to unrelated firms for years. The paper shows that fiscal policy can generate a powerful credit-supply shock in crises, and that recapitalization mitigates this amplification.